5.2 Properties of Rational Exponents and Radicals

Name _____ Period ____ Date _____

Algebra II 5.2 · Student sheet

I can…

- simplify expressions with rational exponents.
- write radical expressions in simplest form.
- add, subtract, multiply, and divide radical expressions.

Vocabulary · fill in the missing word

Simplest form of a radical — no perfect nth-power factors in the radicand, no fractions in the radicand, and no radicals in a

_____

Rationalizing the denominator — _____ a radical from the denominator

Conjugates — _____

Like radicals — radicals with the same _____ and the same radicand

Part 1 · True or false?

Use a calculator to test each statement. Mark it true or false. For each true one, name the property.

Statement	True or false?	Property
2^(1/2) · 2^(3/2) = 2^2		
$(5^{(1/3)})^3 = 5$		
$\sqrt{2} \cdot \sqrt{8} = \sqrt{16}$		
$\sqrt{2} + \sqrt{3} = \sqrt{5}$		
$\sqrt[3]{54} = 3\sqrt[3]{2}$		
$\sqrt{9} / \sqrt{4} = \sqrt{(9/4)}$		

Roots can be multiplied and divided, but they can't be _____

Part 2 · Notes: fill in the blanks

$a^m \cdot a^n$ = _____ $(a^m)^n$ = _____ a^m/a^n = _____

$\sqrt[n]{(a \cdot b)}$ = _____ $\sqrt[n]{(a/b)}$ = _____

A radical is in **simplest form** when the radicand has no perfect nth-power _____, there is no _____ in the radicand, and there is no radical in the _____.

The **conjugate** of $a + b\sqrt{c}$ is _____.

Like radicals have the same _____ and the same _____.

Part 3 · Examples (work them with the class)

Example 1

Simplify each expression.

a. $5^{\left(\frac{1}{3}\right)} \cdot 5^{\left(\frac{5}{3}\right)}$ b. $4^{\frac{\frac{5}{2}}{4}}{}^{\left(\frac{1}{2}\right)}$ c. $\left(3^{\left(\frac{2}{3}\right)}\right)^3$

Example 2

Simplify each expression.

a. $\sqrt[3]{16} \cdot \sqrt[3]{4}$ b. $\sqrt[4]{80} / \sqrt[4]{5}$

Example 3

Write each expression in simplest form.

a. $\sqrt[5]{64}$ b. $\sqrt[4]{\left(\frac{5}{4}\right)}$

Example 4

Write $\frac{1}{5+\sqrt{3}}$ in simplest form.

Example 5

Simplify each expression.

a. $7 \cdot \sqrt[5]{12} - \sqrt[5]{12}$ b. $\sqrt[3]{54} + \sqrt[3]{2}$

Example 6

Simplify each expression. Assume all variables are positive.

a. $\sqrt[3]{(64y^6)}$ b. $\sqrt[4]{(\frac{x^4}{y^8})}$

Part 4 · Partner practice

Partner 1

Write $\sqrt[3]{250}$ in simplest form.

Partner 2

Write $\frac{3}{\sqrt{7} - 2}$ in simplest form.